

Name:

Exam Style Questions

Bearings



Equipment needed: Pen, Pencil, Protractor, Ruler

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

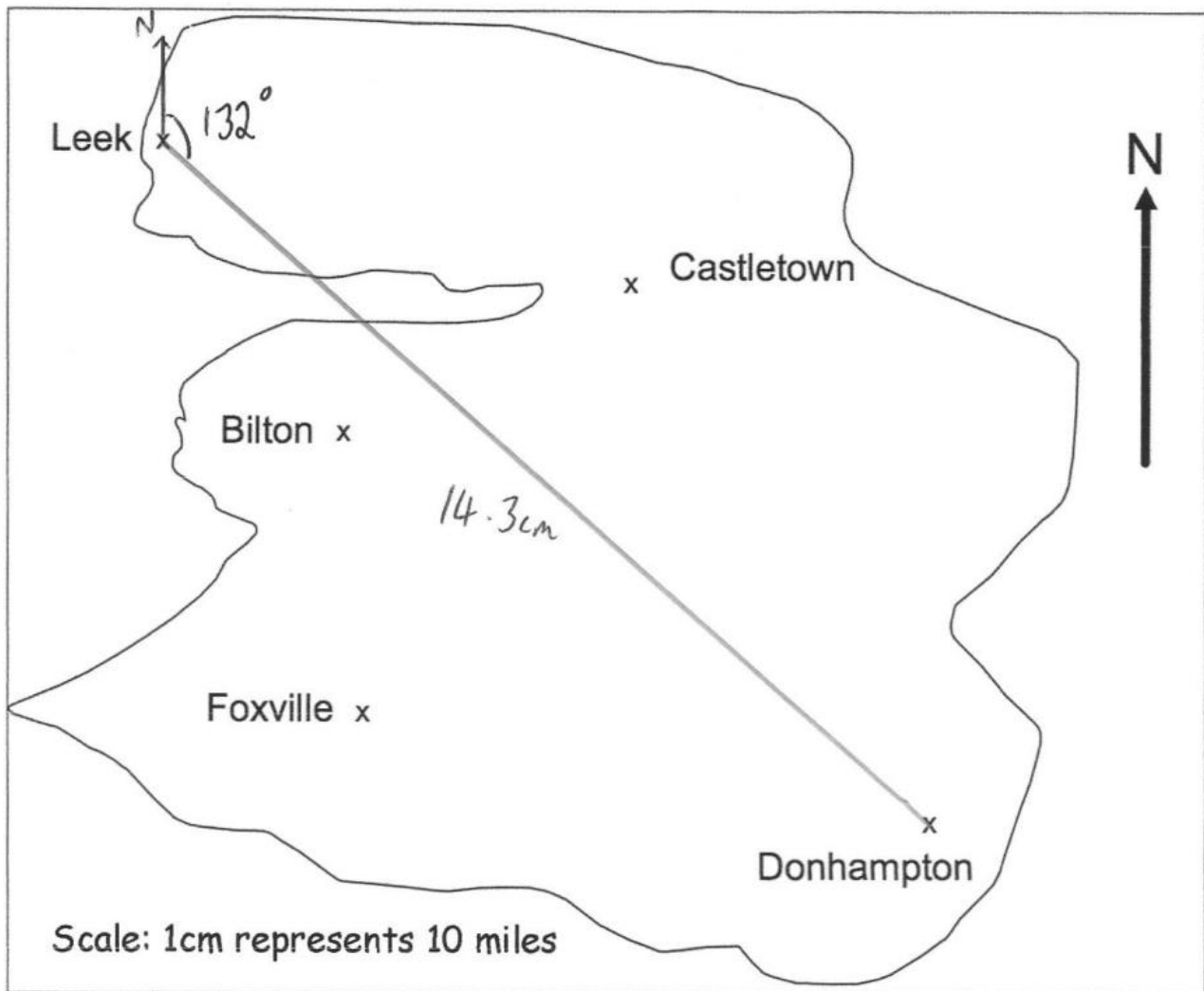
Video 26



Answers and Video Solutions



1. This is a map of an island.



A helicopter flies in a straight line from Leek to Donhampton.

- (a) How far does the helicopter fly?

$$14.3 \times 10 = 143$$

* may depend on printing size.

143miles
(2)

- (b) Write down the bearing of Donhampton from Leek.

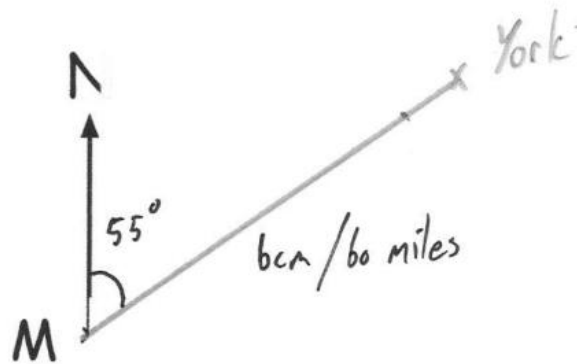
* This may vary due to printing.

132°°
(1)

2. The diagram shows the position of Manchester.



Scale: 1cm represents 10 miles



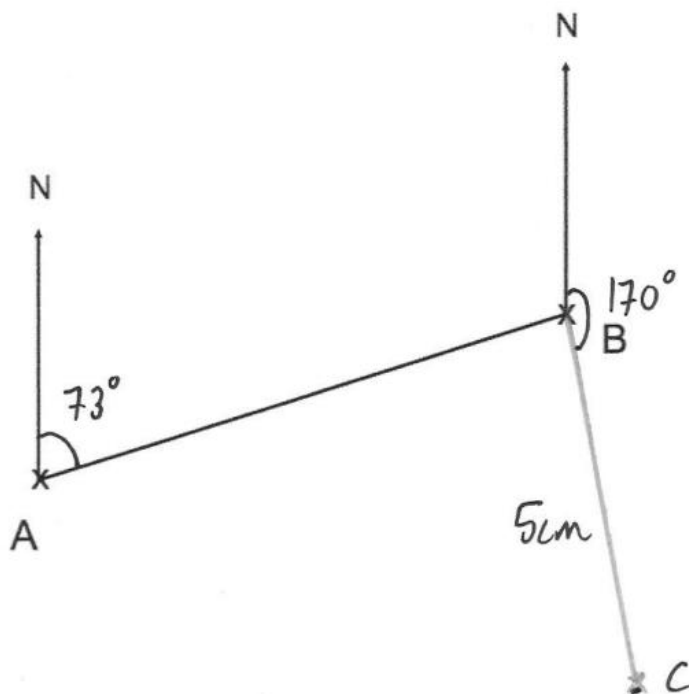
6cm

York is 60 miles away from Manchester on a bearing of 055°

Mark the position of York on the diagram.

(2)

3. The diagram shows the position of two houses, A and B, on a map.



- (a) Measure the bearing of B from A.

* may vary due to printing.

073°
(1)

Another house C is on a bearing of 170° from B.
On the map, C is 5 cm from B

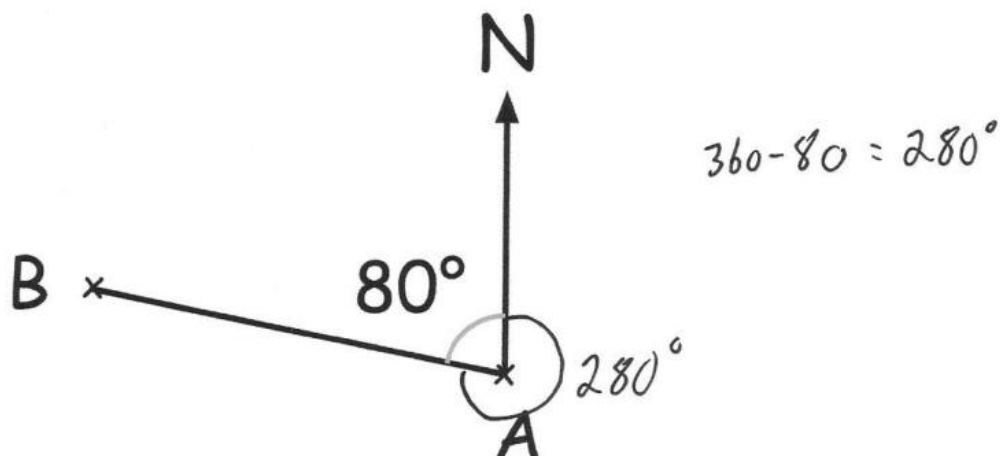
- (b) Mark the position of C with a cross (x) and label it C.

(2)

4.



Olivia has been asked to find the bearing of B from A.
Shown below is her method.



Olivia's answer is 080°

Explain Olivia's mistake.

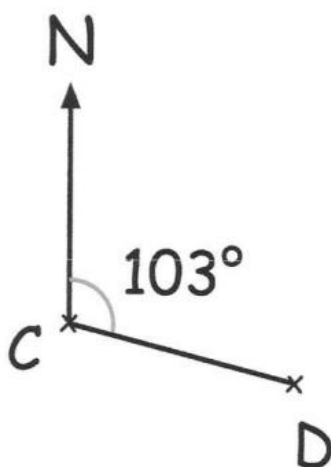
Olivia has measured the anticlockwise angle.
The correct answer is 280°.

(2)

5.



Oliver has been asked to find the bearing of C from D.
Shown below is his method.



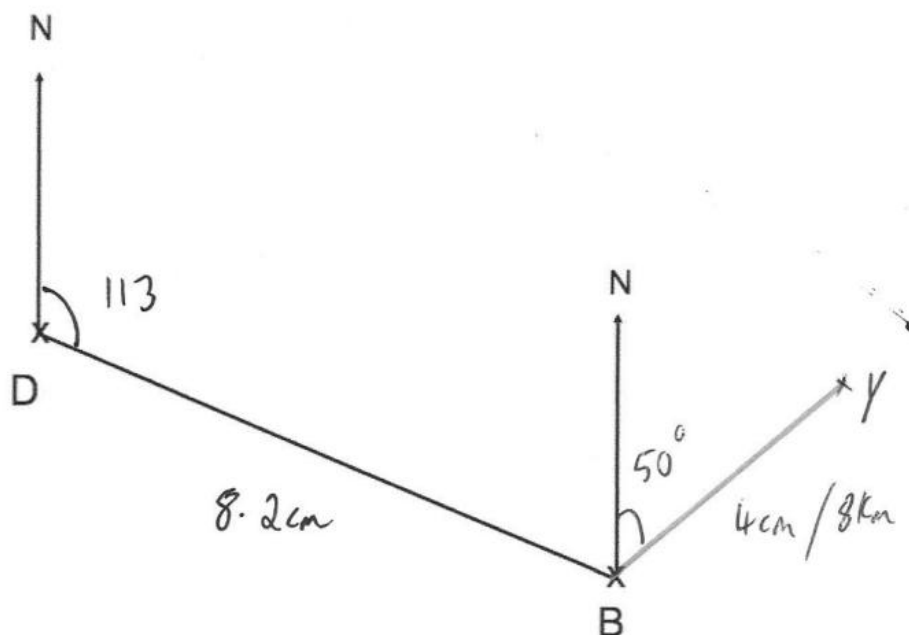
Olivia's answer is 103°

Explain Olivia's mistake.

Olivia has measured the bearing of D from C,
not C from D.
That answer would be 283°.

(2)

6. The diagram shows the position of a boat B and a dock D.



The scale of the diagram is 1 cm represents 2 km.

- (a) Work out the actual distance between the dock and the boat.

$$8.2 \times 2 = 16.4$$

16.4
..... km
(2)

* may vary due to printing.

- (b) Measure the bearing of the boat B from the dock D.

* may vary due to printing.

113
..... °
(1)

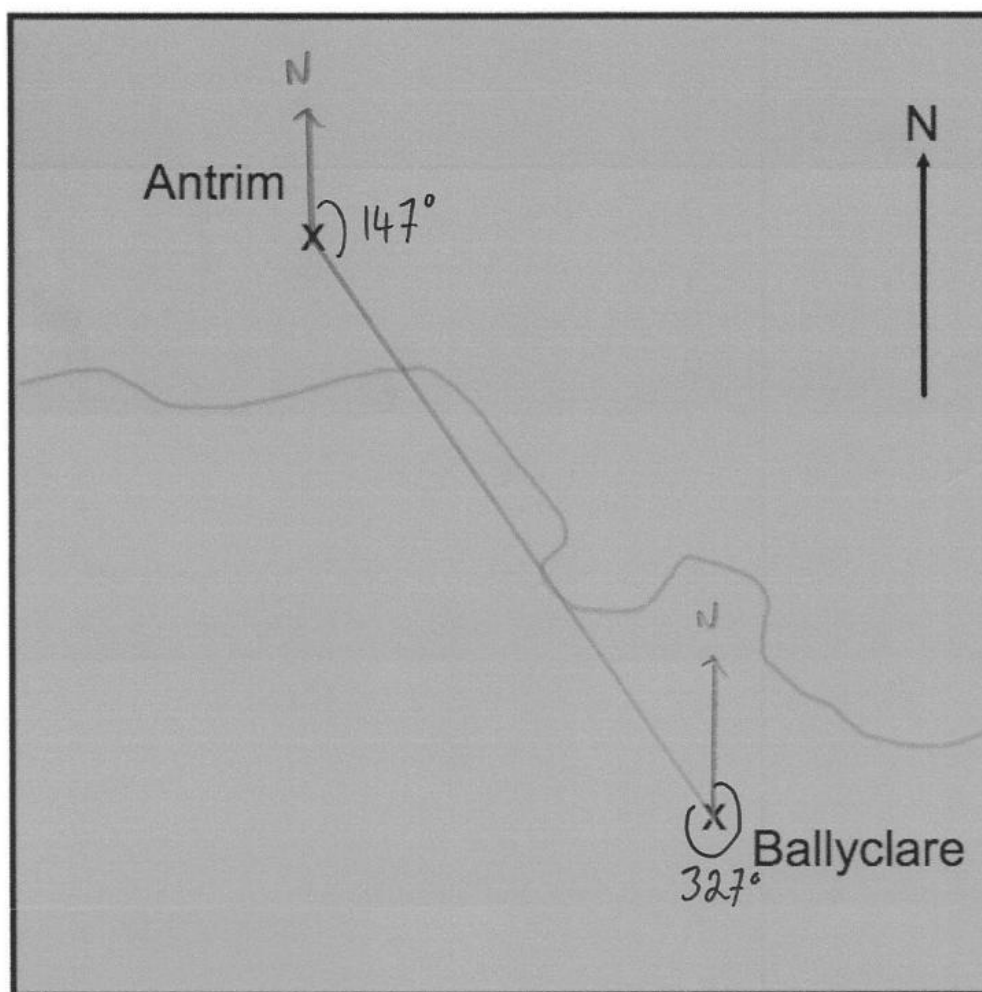
A yacht Y is 8 km from the boat B on a bearing of 050°

- (c) On the diagram, mark the position of yacht Y with a cross (x).
Label it Y.

$$8 \div 2 = 4 \text{ cm}$$

(2)

7. The map below shows the position of two towns.



- (a) Find the bearing of Ballyclare from Antrim.

* This may vary due to printing.

147°
(1)

- (b) Find the bearing of Antrim from Ballyclare.

* This may vary due to printing.

327°
(1)

8.

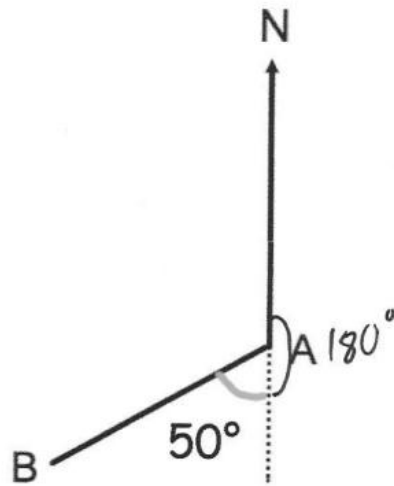


Diagram not drawn accurately

Work out the bearing of B from A.

$$180 + 50 = 230^\circ$$

230

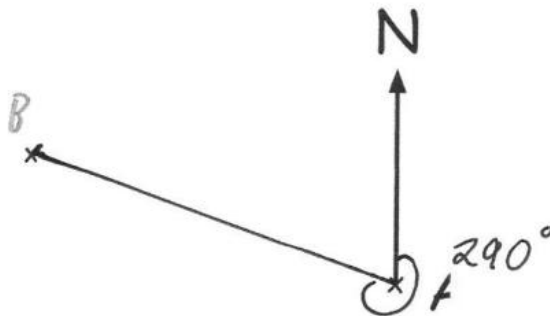
(2)

9.

Sam stands at point A.



Scale: 1cm represents 20m



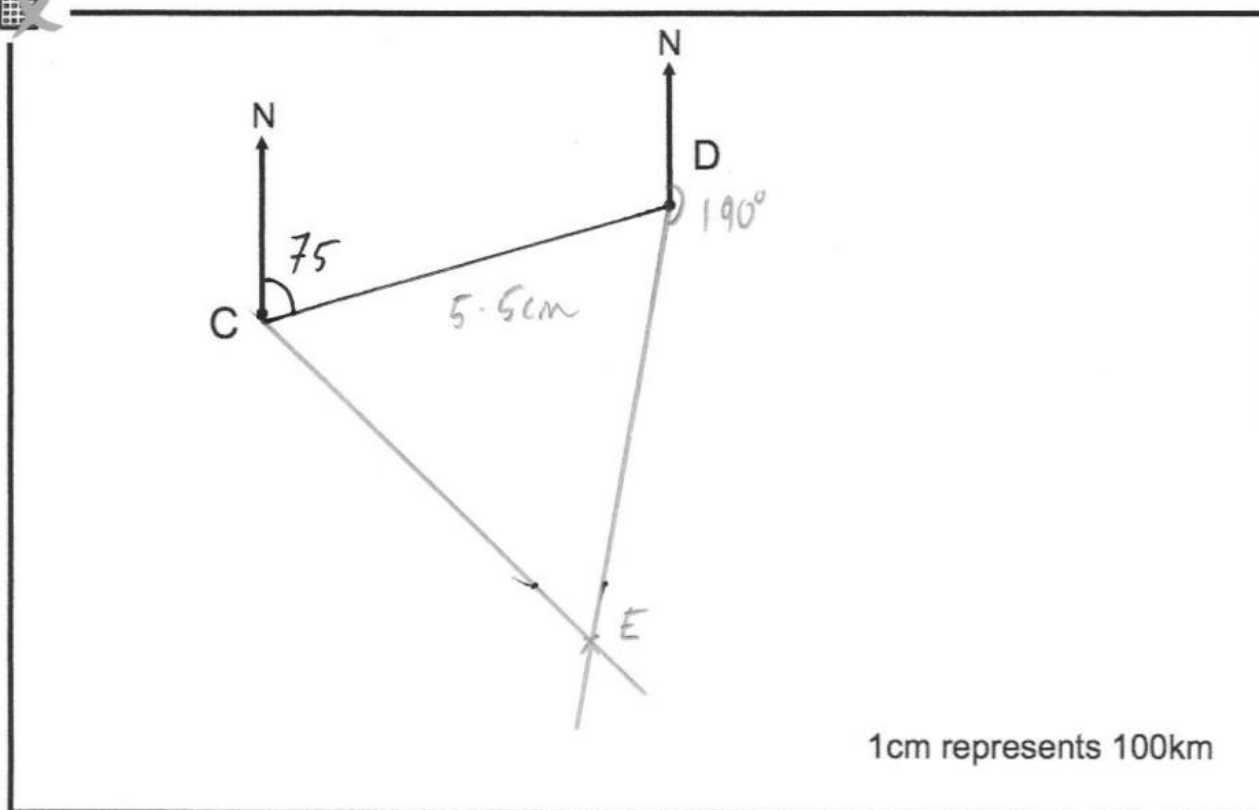
$$100 \div 20 = 5 \text{ cm}$$

Sam runs 100m on a bearing of 290°

Mark Sam's finishing position, B, with a cross.

(2)

10. The diagram shows the position of two cities C and D.



- (a) Work out the actual distance of D from C.

$$5.5 \times 100 = 550$$

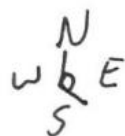
550 km
(2)

- (b) Find the three figure bearing of D from C.

075°
(1)

E is South-East of C.

- (c) Write down the bearing of E from C.



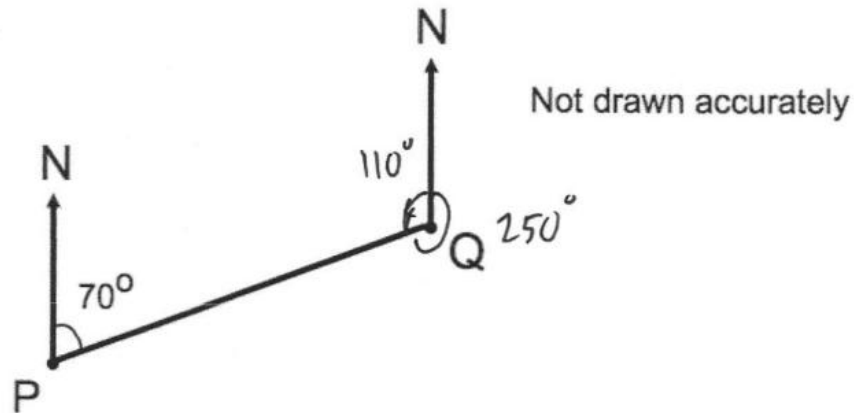
135°
(1)

E is also on a bearing of 190° from D.

- (d) Mark the position of E on the diagram.

(2)

11. The diagram shows the position of two airplanes, P and Q.



The bearing of Q from P is 070°

Calculate the bearing of P from Q.

250

(2)

12. The bearing of C from D is 165°



Calculate the bearing of D from C.

$$165 + 180 = 345$$

345

(2)

13. The bearing of F from G is 300°



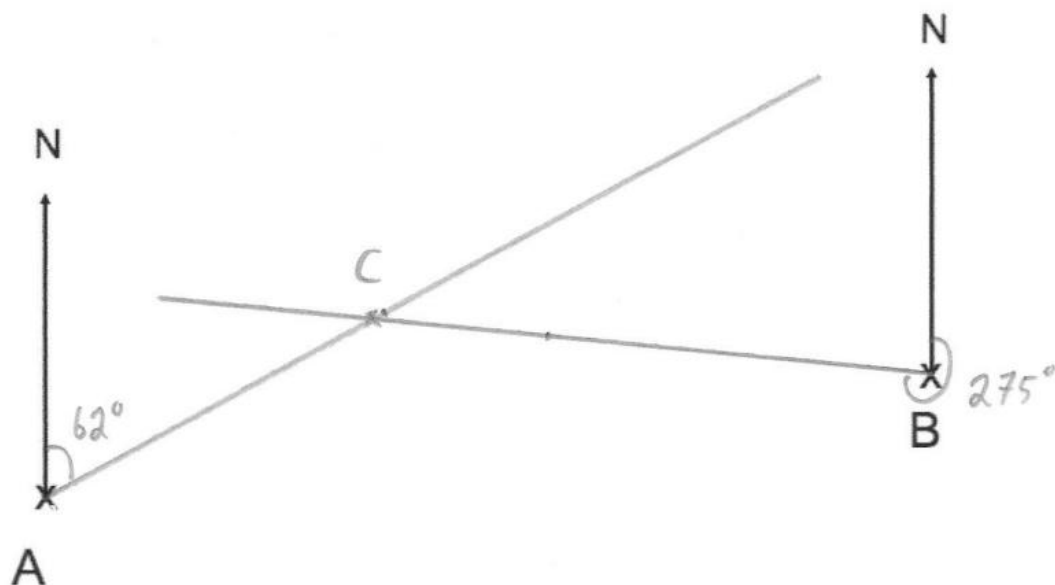
Calculate the bearing of G from F.

$$300 - 180 = 120$$

120

(2)

14. The diagram shows the position of two people, A and B, who are on their Duke of Edinburgh expedition.



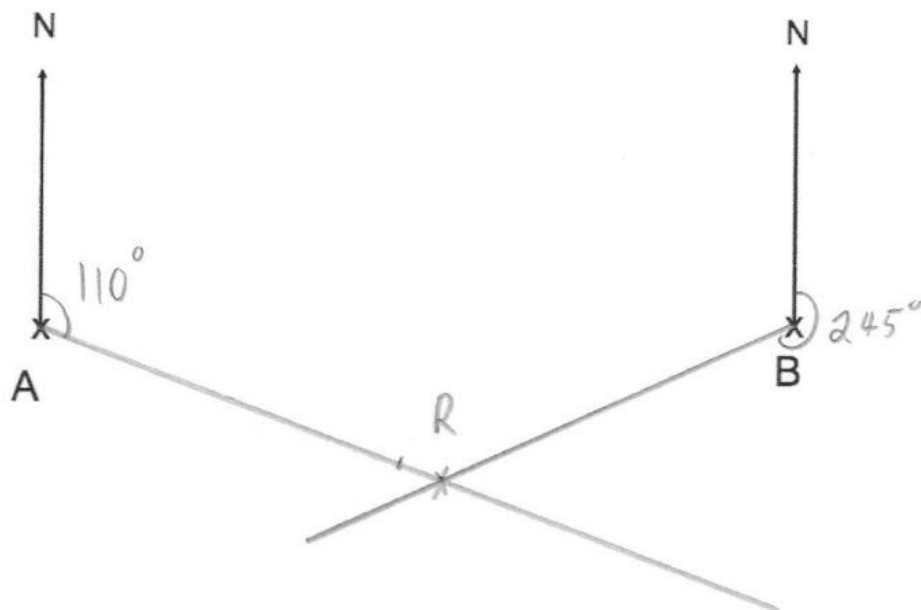
The bearing of person C from person A is 062°

The bearing of person C from person B is 275°

In the space above, mark the position of person C with a cross (x). Label it C.

(3)

15. The diagram shows the position of two towns, A and B.

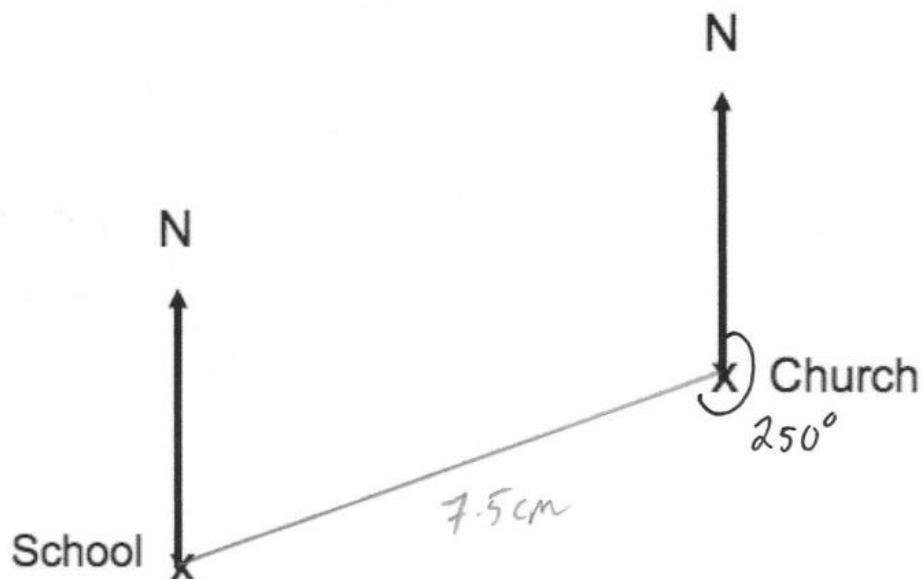


A rugby club, R, has bearing of 110° from town A.
The rugby club, R, has bearing 245° from town B.

In the space above, show the position of the rugby club R.
Mark the position with a cross (x) and label it R.

(3)

16. The map below shows the position of a church and a school.



The scale of the map is 1 : 10,000

- (a) Find the actual distance between the church and school.
Give your answer in metres.

* May vary due to printing size.

$$7.5 \times 10000 = 75000 \text{ cm}$$

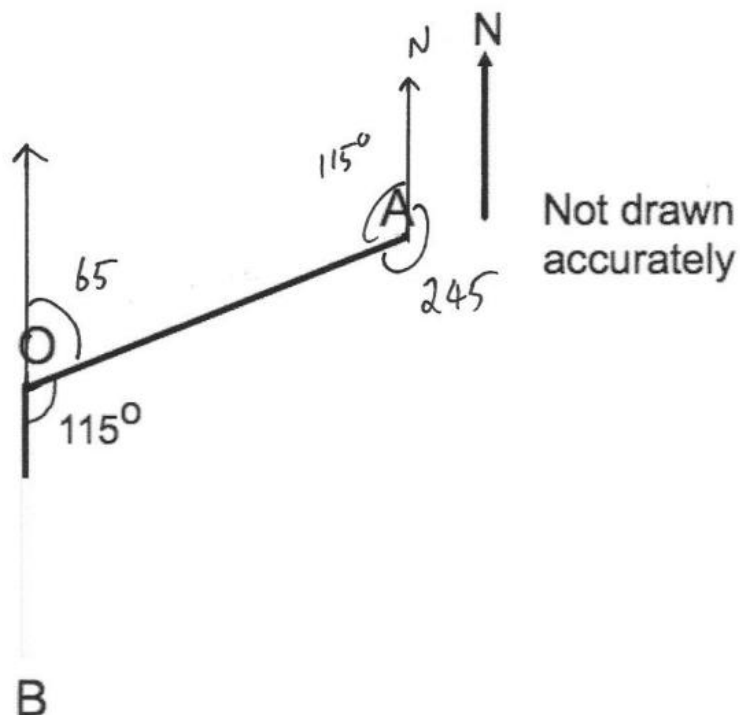
$$75000 \div 100 = 750 \text{ m}$$

750
.....m
(2)

- (b) Find the bearing of the school from the church.

250
.....°
(2)

17. Gregory is at O and there are two roads, one towards A and another towards B. B is due South of O.



Gregory walks towards A.

- (a) On what bearing does he walk?

065°
.....°
(2)

Joshua is at A and walks towards Gregory.

- (b) On what bearing does he walk?

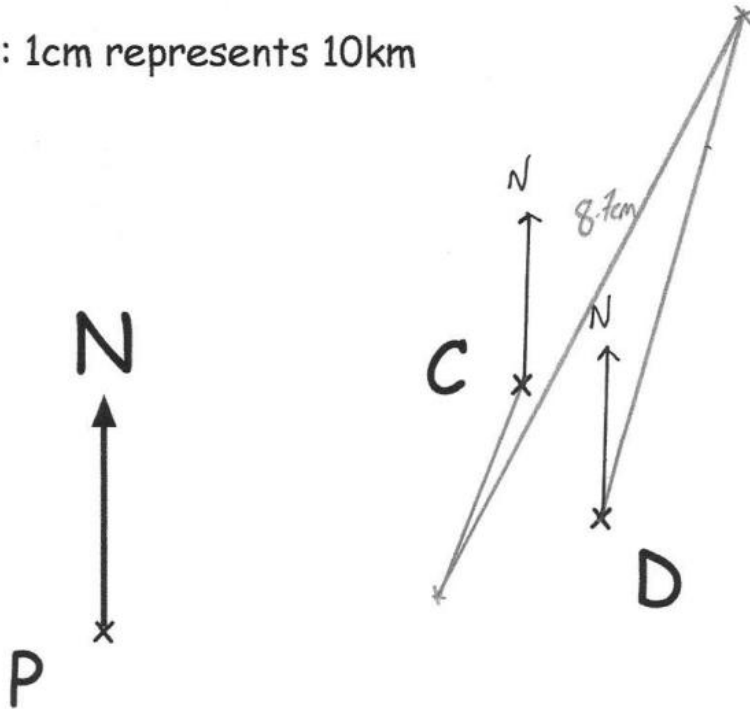
$$360 - 115 = 245$$

245°
.....°
(2)

18. The diagram shows the position of a port P and two boats, C and D at 6am.



Scale: 1cm represents 10km



Boat C sails on a course of 200° at a speed of 15km/h $15 \times 2 = 30 \text{ km}$ 3 cm
 Boat D sails on a course of 015° at a speed of 35km/h $35 \times 2 = 70 \text{ km}$ 7 cm

Find how far apart the boats are at 8am.
 Give your answer in kilometres.

2 hours

8.7 cm

87 km

87 km
 (4)